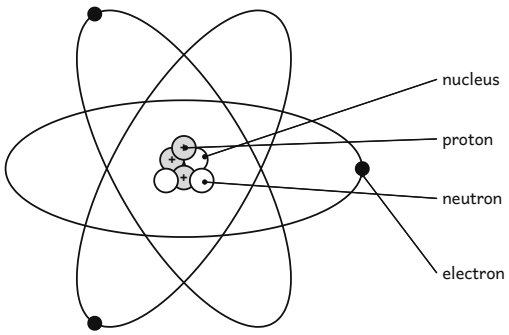
PHYSICAL AND CHEMICAL SCIENCE

# Matter, Atoms and Conservation

Atoms and elements, conservation of mass in open and closed systems, and nuclear changes




**In this pack**

**Learn it:** Atoms: nothing is lost, everything changes ★☆☆
**Fact check:** Atoms and mass ★☆☆
**Think it through:** Where did the mass go? ★★☆
**Investigate:** Is mass conserved? ★★☆
**Real-world case study:** Energy from atoms: Rooppur ★★★
**Review and challenge:** Show what you know ★★★

**This pack belongs to:** _____ **Class:** _____

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LEARN IT · C3 · ATOMS AND CONSERVATION

# Atoms: nothing is lost, everything changes

★☆☆
Grade 5

Name: Class: Date:

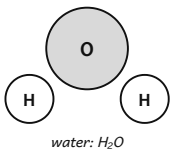
**Learning goals**

- Explain that matter is made of atoms and that atoms are conserved in physical and chemical changes.
- Explain why mass stays the same in a closed system.
- Describe simply how nuclear changes are different.

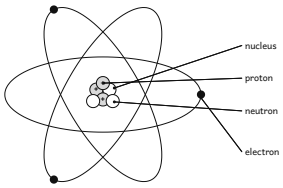
Everything around you (air, water, your desk, your body, the stars) is made of **matter**, and all matter is made of unbelievably tiny particles called **atoms**. A single human hair is about half a million atoms wide!

## Atoms and elements

There are about 118 different kinds of atom. A material made of only one kind of atom is an **element**: oxygen, carbon, iron, gold and aluminium are elements. Atoms can join together to make **molecules**. A water molecule is two hydrogen atoms joined to one oxygen atom: **$H_2O$**.



*water: $H_2O$*




Each atom has a tiny centre called the **nucleus**, made of **protons** and **neutrons**, with even tinier **electrons** moving around it.

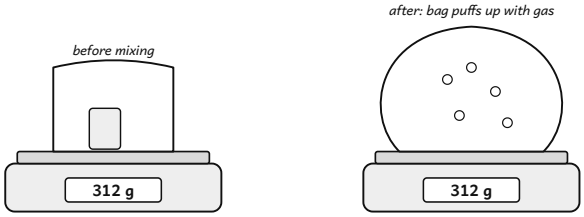
## Atoms are conserved

In a **physical change** (melting, dissolving, evaporating) atoms just move around. In a **chemical change** (burning, rusting, cooking) atoms are **rearranged** to make new substances. But in both kinds of change, **no atoms are made and none are destroyed**. This is the **law of conservation of mass**: the total mass stays the same.

- 100 g of ice melts into 100 g of water.
- 250 g of water + 20 g of salt makes 270 g of salt solution.

## Open and closed systems

Sometimes mass *seems* to disappear. When vinegar and baking soda react in an open cup, the mass goes down. That is because a **gas** (carbon dioxide) escapes into the air. If the reaction happens in a **sealed bag**, a **closed system**, nothing can escape and the mass stays exactly the same.




**From the past**

In the 1780s the French scientist **Antoine Lavoisier** weighed substances very carefully before and after reactions in sealed containers. He showed that mass is never lost, only changed. He is often called the father of modern chemistry.

## Energy is conserved too

Energy cannot be made or destroyed either: it only moves from one place to another or changes form. In a closed system, the total amount of **energy** and **matter** stays the same.

### Nuclear changes: atoms change

In a **nuclear** change, the nucleus itself changes. In a nuclear power station, big uranium atoms are **split** into smaller atoms of different elements, giving out a huge amount of heat. In the Sun, small hydrogen atoms **join** to make helium. So here atoms are **not** conserved: one element turns into another. But if you count all the **protons and neutrons**, the total before and after is the **same**.

| Before | After |
|---|---|
| 1 uranium atom: 236 protons + neutrons | barium (141) + krypton (92) + 3 free neutrons |

141 + 92 + 3 = **236**. Nothing is lost!

**Key words**

| | |
|---|---|
| **matter** | anything that has mass and takes up space |
| **atom** | the tiny building block of matter |
| **element** | a substance made of only one kind of atom |
| **molecule** | atoms joined together |
| **conservation of mass** | mass stays the same in a change |
| **closed system** | nothing can get in or out |
| **nucleus** | the centre of an atom |
| **nuclear change** | a change in the nucleus that turns one element into another |

# Atoms and mass



Name: Class: Date:

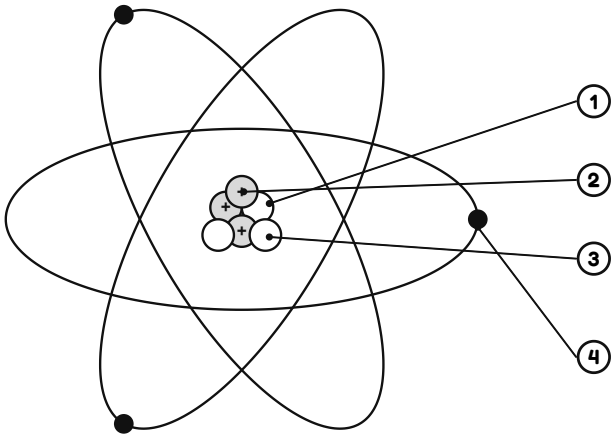
**Learning goals**

- Label an atom.
- Recall the law of conservation of mass.

## 1 Label the atom. [4]

LABELS

**neutron nucleus electron proton**




1 _____ 2 _____

3 _____ 4 _____

## 2 Fill in the gaps. [5]

WORD BANK

**element bigger closed destroyed open mass atoms**

All matter is made of tiny _____. A substance made of only one kind of atom is an _____. In physical and chemical changes, atoms are not made or _____, so the total _____ stays the same. A sealed bag is a _____ system.

## 3 Work out the missing masses. [3]

| Change | Mass before | Mass after |
|---|---|---|
| ice melting | 100 g of ice | |
| dissolving sugar | 200 g water + 15 g sugar | |
| freezing juice | 330 g of juice | |

**4** **True or false?** [4]

| Statement | True | False | If false, correct it |
|---|---|---|---|
| 1. When sugar dissolves, its atoms are destroyed. | | | |
| 2. Water is made of hydrogen and oxygen atoms. | | | |
| 3. In a nuclear change, one element can turn into another. | | | |
| 4. Burning wood makes the atoms disappear. | | | |

# Where did the mass go?

★★☆
Grade 5

Name: Class: Date:

**Learning goals**

- Explain changes in mass using open and closed systems.
- Apply conservation to new examples.

**1** **A candle has a mass of 50 g. After burning for an hour its mass is 42 g. Has mass been destroyed? Explain.** [2]

**2** **Describe an experiment to show that mass is conserved when a candle burns.** [2]

**3** **An iron nail gets heavier as it rusts. Where does the extra mass come from?** [2]

**4** **Use the idea of conservation to check this nuclear change. Are the protons and neutrons conserved?** [2]

| Before | After |
|---|---|
| a nucleus with 235 protons + neutrons, hit by 1 neutron | 139 + 94 + 3 free neutrons |

INVESTIGATE · C3 · ATOMS AND CONSERVATION

# Is mass conserved?

★★☆
Grade 5

Name: Class: Date:

**Learning goals**

- Measure mass accurately.
- Compare open and closed systems.
- Explain unexpected results.

**You will need**

a digital balance, a zip-lock bag, a small cup, vinegar, baking soda, a teaspoon, an open plastic cup

**Stay safe**

Wear goggles. Vinegar can sting your eyes. Point the bag away from faces.

Class 5C carried out three tests and measured the mass before and after.

**1** **Complete the table.** [3]

| Test | Mass before (g) | Mass after (g) | Change (g) |
|---|---|---|---|
| dissolving salt in water (open cup) | 270 | 270 | |
| vinegar + baking soda in a sealed bag | 312 | 312 | |
| vinegar + baking soda in an open cup | 312 | 309 | |

**2** **Why did the mass go down in the open cup but not in the sealed bag?** [2]

**3** **What evidence showed that a gas was made?** [1]

**4** **Write a conclusion about conservation of mass.** [2]

**5** **The balance shows whole grams only. How could this affect the results?** [1]

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REAL-WORLD CASE STUDY · C3 · ATOMS AND CONSERVATION

# Energy from atoms: Rooppur

★★★ Grade 5

Name: Class: Date:

**Learning goals**

- Explain nuclear power simply.
- Weigh advantages and disadvantages.
- Consider different perspectives.

Science · Geography · Economics · Ethics · Environment

Bangladesh needs more and more electricity for homes, schools, hospitals and factories. Most of it comes from burning **natural gas**, which is running out and gives off carbon dioxide.

At **Rooppur**, in Pabna district on the bank of the Padma, Bangladesh is building its **first nuclear power plant**, with help from Russia, and it is starting up in the mid-2020s. Inside its reactors, uranium atoms are split. The heat boils water into steam, which spins turbines to make electricity. A small amount of uranium can produce a very large amount of energy, and the plant gives off almost no carbon dioxide while working.

But nuclear power also has challenges: it is very **expensive** to build; it must be run with great **safety** care; and the used fuel stays **radioactive** (dangerous) for thousands of years, so it must be stored very carefully.

Based on public information about the Rooppur Nuclear Power Plant.

**1** **Put the steps of making electricity at Rooppur in order.** [3]

- ☐ Steam spins a turbine.
- ☐ Uranium atoms split in the reactor.
- ☐ The turbine turns a generator, making electricity.
- ☐ The heat boils water into steam.
- ☐ This releases a lot of heat.

**2** **Complete the table.** [6]

| Advantages of nuclear power | Disadvantages of nuclear power |
|---|---|
| | |
| | |
| | |

**3** **What might each person think about the plant?** [3]

| Person | View |
|---|---|
| a factory owner in Dhaka | |
| a farmer living near Rooppur | |
| a climate scientist | |

**4** **Your opinion: should Bangladesh get more electricity from nuclear power, solar power or both? Give reasons.** [2]

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# Show what you know

★★★
Grade 5

Name: Class: Date:

**Learning goals**

- Use key vocabulary.
- Answer test-style questions.

**1** **Find the key words.**

| A | T | O | M | V | S | V | T | H | J | Y |
|---|---|---|---|---|---|---|---|---|---|---|
| D | A | W | S | T | N | P | I | C | E | G |
| B | O | R | E | F | M | A | S | S | G | P |
| T | F | M | G | L | P | R | O | T | O | N |
| K | S | E | A | L | E | D | T | C | K | E |
| I | F | O | S | Y | T | M | Y | F | U | M |
| O | E | Y | N | U | C | L | E | U | S | D |
| E | L | E | C | T | R | O | N | N | L | L |
| R | H | K | M | J | S | I | A | W | T | A |
| C | M | U | M | O | L | E | C | U | L | E |
| Y | N | E | U | T | R | O | N | W | A | I |

**atom electron element mass molecule neutron nucleus proton sealed**

**2** **Salt (20 g) is dissolved in water (180 g). What is the mass of the solution?** [1]

- ◯ 160 g
- ◯ 180 g
- ◯ 200 g
- ◯ 220 g

**3** **Which is a closed system?** [1]

- ◯ an open pan of boiling water
- ◯ a sealed jar
- ◯ a candle burning in a room
- ◯ a puddle drying

**4** **Draw particle pictures to show that 10 atoms before a change are still 10 atoms after.** [2]

before

after

**5** **Challenge. Explain why scientists say atoms are conserved in chemical changes but not in nuclear changes, and what is still conserved in a nuclear change.** [3]

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**How am I doing? Colour or tick a circle.**

| I can... | Not yet | Nearly | Got it! |
|---|---|---|---|
| explain that matter is made of atoms | ○ | ○ | ○ |
| explain why mass is conserved in physical and chemical changes | ○ | ○ | ○ |
| explain the difference between open and closed systems | ○ | ○ | ○ |
| measure mass before and after a change | ○ | ○ | ○ |
| explain how nuclear changes are different | ○ | ○ | ○ |
| discuss the advantages and disadvantages of nuclear power | ○ | ○ | ○ |